

CHANGELOG

Append-only history of documentation-relevant product changes.

2026-08-02 — Redis cache-aside (Option C, local)

Field	Value
Feature	Document Redis cache-aside how-it-works; mark learning topics current
Files Modified	MODULES/Caching.md (new); TECH_STACK , ARCHITECTURE , TECH_LEARNING_PLAN , README , DEVELOPMENT_GUIDE , DATABASE_SCHEMA , CALL_FLOW , SEQUENCE_DIAGRAMS , CHANGE_IMPACT_GUIDE , FUNCTIONAL_SPECIFICATION , EXTENSION_GUIDE , INTERVIEW_QA , MODULES/Customer , MODULES/Account , CHANGELOG
Classes / APIs covered	com.bankone.cache (RedisCacheConfig , CacheNames , CacheKeys); @Cacheable / @CacheEvict on customer/account/policy/user/role/transaction/beneficiary/dashboard/transfer/report services; APP_CACHE_REDIS
Reason	User asked to update all docs with how caching works
Impact	Docs describe cache-aside flow, TTLs, eviction, JDK serialization, verify steps; prod remains cache-off by default

2026-07-31 — Docs sync + technical learning plan

Field	Value
Feature	Full learning roadmap doc; sync product docs with current code
Files Modified	docs/_source/TECH_LEARNING_PLAN.md (new); README , FUNCTIONAL_SPECIFICATION , API_DOCUMENTATION , DATABASE_SCHEMA , ARCHITECTURE , EXTENSION_GUIDE , TECH_STACK , MODULES/Audit , MODULES/Reports , MODULES/Portal (new), CHANGELOG
Classes / APIs covered	Audit event trail + backfill; portal/beneficiaries/approvals; reports+PDF; multi-role staff; logout audit
Reason	User asked to include all learning topics in the plan and update docs to current functionality
Impact	Docs distinguish implemented vs planned; learning track separate from deferred hardening

2026-07-26 — Interview follow-up answers

Field	Value
Feature	Answer the remaining follow-up interview questions
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User asked for direct answers to the end-of-guide follow-up prompts
Impact	The handbook now includes concrete answers for JWT, outbox, audit, branch permissions, and duplicates

2026-07-26 — Interview wording polish pass

Field	Value
Feature	Smooth remaining awkward sentences in the handbook
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User asked for a re-check of phrasing correctness
Impact	A few note-like lines were rewritten to sound more natural

2026-07-26 — Interview handbook prose polish

Field	Value
Feature	Polish handbook wording for more natural interview phrasing
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User asked whether the sentences were phrased correctly
Impact	Tone is now less template-like and more interview-ready

2026-07-26 — Interview follow-up drills and revision sheet

Field	Value
Feature	Add follow-up drill answers and a 5-minute revision sheet
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User wanted deeper interview prep with memorisable answers
Impact	Handbook now has a revision summary and topic-specific drill questions

2026-07-26 — Interview handbook framework

Field	Value
Feature	Add answer framework, traps, and topic playbooks to interview guide
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User wanted deeper memorisable interview prep, not just Q&A bullets
Impact	Guide now teaches how to answer, what to avoid, and how the flows work

2026-07-26 — Interview guide deep-dive walkthroughs

Field	Value
Feature	Add long-form flow explanations for login, onboarding, accounts, Kafka, frontend, security, and deployment
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User asked for deeper interview answers rather than one-line notes
Impact	The guide now teaches end-to-end reasoning, not just surface definitions

2026-07-26 — Interview Q&A expanded again

Field	Value
Feature	Add deeper backend, frontend, deployment, and design tradeoff interview questions
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/CHANGELOG.md
Reason	User asked for a much more exhaustive Q&A set
Impact	The interview guide now covers ~160 questions and deeper technical follow-ups

2026-07-26 — Extensive interview Q&A doc

Field	Value
Feature	Expand INTERVIEW_QA.md with codebase-specific technical and functional questions
Files Modified	docs/_source/INTERVIEW_QA.md , docs/_source/README.md , docs/_source/CHANGELOG.md
Reason	User requested an extensive interview prep list grounded in the current codebase
Impact	Docs index now links to the interview guide; the guide covers backend, frontend, security, Kafka, and deployment

2026-07-25 — Docs sync: Kafka notifications, pagination, load-test seed

Field	Value
Feature	Documentation catch-up for Kafka→email, list pagination jump, Liberty JDWP, load-test SQL
Files Modified	<code>docs/_source/*</code> , <code>docs/DEPLOY.md</code> , <code>MODULES/Notification.md</code> , regenerate DOCX/PDF
Reason	Code landed for notifications/pagination/debug/seed; docs lagged
Impact	Staff docs match implemented stack (Aiven + SendGrid HTTP, Mailpit local)

2026-07-24 — Kafka notifications → customer email

Field	Value
Feature	After account open/deposit/withdraw/transfer, publish <code>BankActionEvent</code> to Kafka; consumer emails customer
Files Modified	<code>com.bankone.notification.*</code> ; <code>AccountServiceImpl</code> ; <code>application.properties</code> ; <code>docker-compose.yml</code> (kafka, mailpit); <code>.env.example</code>
Classes / Methods	<code>NotificationEventPublisher.publish</code> ; <code>NotificationEventConsumer.onMessage</code> ; <code>SmtplibNotificationMailService</code> / <code>SendGridNotificationMailService</code> ; <code>NotificationEmailComposer</code>
Reason	Async customer notifications; learn Kafka; Render SMTP blocked → SendGrid HTTPS
Impact	Local Mailpit :8025 ; Render needs Aiven + <code>MAIL_TRANSPORT=sendgrid</code> ; recipient = <code>customers.email</code>

2026-07-24 — List pagination First/Last/Go to page

Field	Value
Feature	Shared <code>list-pagination</code> : first/last + jump-to-page (1-based UI → 0-based API)
Files Modified	<code>list-pagination.ts/html/scss/spec.ts</code>
Reason	10k+ load-test rows made Prev/Next-only unusable
Impact	Accounts, customers, employees, nested account/tx lists

2026-07-24 — Load-test seed 10k customers

Field	Value
Feature	<code>scripts/seed-loadtest-10k.sql</code> — 10k LoadTest customers + 12k accounts (branch 9999)
Reason	Pagination / list performance testing
Impact	Re-runnable; identifiable emails <code>loadtest.customer*@bankone.test</code>

2026-07-24 — Liberty redeploy always JDWP debug

Field	Value
Feature	<code>scripts/redeploy-liberty.sh</code> writes JDWP <code>suspend=n</code> on port 7777 by default
Reason	Attach IntelliJ Remote Debug after every redeploy
Impact	<code>LIBERTY_DEBUG=0</code> disables; <code>WLP_DEBUG_ADDRESS</code> overrides port

2026-07-24 — Fix Liberty boot after Jackson property

Field	Value
Feature	Remove invalid <code>spring.jackson.serialization.write-dates-as-timestamps</code>
Files Modified	<code>application.properties</code>
Reason	Spring Boot 4 / Jackson 3 rejects that property → app failed to start on Liberty → login “Failed to fetch”
Impact	Redeploy WAR; Instant still emits UTC Z without that flag

2026-07-24 — Transaction timestamps UTC

Field	Value
Feature	Ledger <code>createdAt</code> as Instant (UTC/ Z); UI <code>asUtc</code> + local <code>date</code> pipe
Files Modified	<code>Transaction.java</code> , <code>TransactionResponse.java</code> , <code>application.properties</code> ; <code>as-utc.pipe.ts</code> , <code>account-detail</code>
Reason	Render UTC vs browser IST made debit/credit times look ~5.5h off
Impact	New/updated ledger rows align with header clock in IST; legacy rows treated as UTC via <code>asUtc</code>

2026-07-24 — Transfer between accounts

Field	Value
Feature	<code>POST /accounts/{fromId}/transfer</code> ; Accounts Transfer dialog
Files Modified	<code>TransferRequest</code> , <code>AccountService(Impl).transfer</code> , <code>AccountController</code> ; FE <code>account.ts</code> , <code>transfer-dialog</code> , <code>account-list</code>
Methods Modified	<code>transfer</code> (ordered pessimistic locks; DEBIT+CREDIT ledger)
Reason	Slice 5 — move money between two ACTIVE same-currency accounts
Impact	UI From/To pickers on Accounts; no beneficiaries yet

2026-07-24 — Withdraw (DEBIT) + UI

Field	Value
Feature	POST /accounts/{id}/withdraw ; Accounts Withdraw dialog
Files Modified	WithdrawRequest , AccountRepository.findByIdForUpdate , AccountService(Impl).withdraw , AccountController ; FE account.ts , withdraw-dialog , account-list
Methods Modified	withdraw ; lock on withdraw load
Reason	Slice 4 — debit balance with insufficient-funds check + row lock
Impact	Ledger DEBIT “Withdrawal”; UI on Accounts list

2026-07-24 — System-wide loading spinner

Field	Value
Feature	Shared app-loading-state (Material spinner) on list/detail/profile loads
Files Modified	shared/components/loading-state/* ; account/customer/employee lists + details; profile
Reason	Replace plain “Loading...” text with consistent spinner UX
Impact	Dashboard refresh control unchanged; error/empty states still text

2026-07-23 — Netlify npm peer + publish path

Field	Value
Feature	Align Angular 22.0.8 packages; root netlify.toml publish path
Files Modified	package.json , package-lock.json , netlify.toml , docs/DEPLOY.md
Reason	ERESOLVE (animations 22.0.8 vs core 22.0.7); doubled publish path
Impact	npm ci succeeds on Netlify; publish resolves to one BankOne-Frontend/dist/...

2026-07-23 — Netlify frontend deploy fix

Field	Value
Feature	Fix production Angular build + Netlify SPA publish
Files Modified	<code>api-config.ts</code> , <code>angular.json</code> , <code>BankOne-Frontend/netlify.toml</code> , <code>docs/DEPLOY.md</code>
Classes Modified	—
Methods Modified	—
Reason	Wrong <code>environment</code> import broke Netlify build; empty/wrong publish → Netlify 404
Impact	<code>npm run build --configuration=production</code> succeeds; publish <code>dist/BankOne-Frontend/browser</code>

2026-07-23 — Opening deposit writes CREDIT ledger

Field	Value
Feature	openAccount records CREDIT when openingDeposit > 0 (customer create + add account)
Files Modified	<code>AccountServiceImpl.openAccount</code>
Classes Modified	<code>AccountServiceImpl</code>
Methods Modified	<code>openAccount</code>
Reason	History missing for opening deposits
Impact	bank_transaction rows with narration “Opening deposit”

2026-07-23 — Fix blank screen after login (breadcrumbs)

Field	Value
Feature	Make Breadcrumbs resilient so MainLayout does not crash post-login
Files Modified	BankOne- Frontend/src/app/shared/components/breadcrumbs/breadcrumbs.ts
Classes Modified	Breadcrumbs
Methods Modified	buildCrumbs, urlOf, resolveLabel; crumbs toSignal initialValue
Reason	toSignal({ initialValue: this.buildCrumbs() }) ran before router tree ready; pathFromRoot / snapshot.url could throw and blank the app
Impact	After login, MainLayout renders; breadcrumbs fall back to Dashboard on error

—|— —| | **Feature** | GET /accounts/{accountId} and GET /accounts/{accountId}/transactions || **Files Modified** | AccountController.java, AccountService / Impl, TransactionService / Impl, TransactionResponse.java; docs _source API_DOCUMENTATION, MODULES/Account, MODULES/Transaction, CALL_FLOW, FUNCTIONAL_SPECIFICATION, CHANGELOG || **Classes Modified** | AccountController, AccountService / Impl, TransactionService / Impl, TransactionResponse || **Methods Modified** | getAccountById, getTransactions, TransactionService.getByAccountId || **Reason** | Staff can load one account and its paged ledger without UI yet || **Impact** | List API implemented (Partial overall: no withdraw/transfer/UI; dashboard count still stub) |

2026-07-23 — Technology stack living document

Field	Value
Feature	Add <code>TECH_STACK</code> inventory; keep in sync on dependency changes
Files Modified	<code>docs/_source/TECH_STACK.md</code> , <code>docs/_source/README.md</code> , <code>.cursor/rules/documentation-architect.mdc</code> , <code>CHANGELOG.md</code>
Classes Modified	N/A
Methods Modified	N/A
Reason	Single place listing all technologies/versions used in the project
Impact	Architect rule requires updating <code>TECH_STACK</code> when <code>pom/package.json/deploy/auth/DB</code> change

2026-07-23 — Transaction ledger foundation + deposit CREDIT write

Field	Value
Feature	Transaction module foundation; deposit posts CREDIT ledger row
Files Modified	<code>transaction/entity/Transaction.java</code> , <code>enums/TransactionType.java</code> , <code>repository/TransactionRepository.java</code> , <code>service/TransactionService.java</code> , <code>TransactionServiceImpl.java</code> ; <code>AccountServiceImpl.java</code> ; <code>docs _source</code> <code>MODULES/Transaction</code> , <code>Account</code> , <code>FUNCTIONAL_SPECIFICATION</code> , <code>CALL_FLOW</code> , <code>DATABASE_SCHEMA</code> , <code>CHANGELOG</code>
Classes Modified	<code>Transaction</code> , <code>TransactionType</code> , <code>TransactionRepository</code> , <code>TransactionService / Impl</code> , <code>AccountServiceImpl</code>
Methods Modified	<code>TransactionServiceImpl.record()</code> ; <code>AccountServiceImpl.deposit()</code>
Reason	Start immutable money-movement history; deposit must not only mutate balances
Impact	<code>bank_transaction</code> table; list/API still stub; dashboard <code>todayTransactionCount</code> still 0; <code>openAccount</code> does not yet write CREDIT

2026-07-23 — Rename BanOne-BackEnd → BankOne-BackEnd

Field	Value
Feature	Fix misspelled backend folder name
Files Modified	Folder rename; scripts, README, docs, architect rule
Classes Modified	N/A
Methods Modified	N/A
Reason	Correct spelling to match BankOne branding
Impact	Re-open IntelliJ on <code>BankOne-BackEnd</code>

2026-07-23 — Move backend into BankOne-BackEnd

Field	Value
Feature	Relocate Maven <code>src + pom.xml</code> under <code>BankOne-BackEnd/</code>
Files Modified	<code>BankOne-BackEnd/**</code> , <code>scripts/redeploy-liberty.sh</code> , <code>README.md</code> , docs paths, architect rule
Classes Modified	N/A (path move only)
Methods Modified	N/A
Reason	Separate backend folder alongside <code>BankOne-Frontend</code>
Impact	Open IntelliJ on <code>BankOne-BackEnd</code> ; redeploy script builds from new root

2026-07-23 — Documentation formatting / alignment fix

Field	Value
Feature	Restyle DOCX + PDF (margins, tables, headings, code blocks)
Files Modified	<code>BankOne/docs/**/*.docx</code> , <code>**/*.pdf</code> , <code>docs/_assets/*</code> , <code>scripts/regenerate-docs.sh</code> , <code>scripts/_polish_docx.py</code>
Classes Modified	N/A
Methods Modified	N/A
Reason	User reported poor alignment inside exported docs
Impact	Open refreshed PDF/DOCX; use regenerate script after content updates

2026-07-23 — Docs format: Word + PDF in project folder

Field	Value
Feature	Replace Markdown docs with DOCX/PDF; move under BankOne/docs
Files Modified	BankOne/docs/** (docx+pdf), removed workspace /docs markdown; .cursor/rules/documentation-architect.mdc ; BankOne/README.md
Classes Modified	N/A
Methods Modified	N/A
Reason	User requested PDF/Word instead of MD, located in the project folder
Impact	Architect rule treats .docx as editable source and .pdf as distribution copy

2026-07-23 — Documentation baseline (Software Architect)

Field	Value
Feature	Establish living /docs synchronized with codebase
Files Modified	docs/**, .cursor/rules/documentation-architect.mdc
Classes Modified	N/A (docs only)
Methods Modified	N/A
Reason	Permanent architect mandate: docs are part of source; capture current implemented state
Impact	All future features must update these docs + append this changelog

2026-07-23 — Account policies GET + opening deposit UX

Field	Value
Feature	Policy-aware opening deposit for Current accounts
Files Modified	Backend: AccountPolicyController , AccountPolicyService / Impl ; Frontend: account-policy.ts , opening-deposit-dialog/* , customer-detail.ts
Classes Modified	AccountPolicyController , AccountPolicyServiceImpl , Angular OpeningDepositDialog , CustomerDetail
Methods Modified	getActivePolicy() , addCurrentAccount() , dialog submit → openAccount
Reason	CURRENT/INR policy requires min opening deposit (seed 5000); UI must not hardcode 1
Impact	Customer detail Current flow; account open validation; docs MODULES/Account.md , API, sequences

2026-07-22 — Accounts list + deposit API

Field	Value
Feature	Accounts search UI + POST /accounts/{id}/deposit
Files Modified	AccountController , AccountServiceImpl , DepositRequest , frontend features/accounts/* , sidebar route
Classes Modified	AccountController , AccountServiceImpl , AccountList
Methods Modified	searchAccounts() , deposit()
Reason	Staff can find accounts and credit balances
Impact	No ledger rows yet; Transaction module still stub

2026-07-23 — System-wide breadcrumbs

Field	Value
Feature	Breadcrumbs in main layout from route <code>data.breadcrumb</code> / <code>data.breadcrumbParents</code>
Files Modified	Frontend: <code>shared/breadcrumbs/*</code> , <code>layout/main-layout/*</code> , feature <code>*.routes.ts</code> (breadcrumb metadata)
Classes Modified	<code>Breadcrumbs</code> , <code>MainLayout</code>
Methods Modified	Breadcrumb build from <code>ActivatedRoute</code> snapshot / router events
Reason	Staff navigation context across Customers, Accounts, Users, Dashboard
Impact	All authenticated shell pages show trail; leaf routes declare parents for nested screens